REUSABLE MARKET SPECIFICATION

LLC Portfolio Identification and Mailing Research

Process requirements, decision rules, data model, and operating diagrams

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MASTER SPECIFICATION — APPLIES TO EVERY MSA

Version 1.0 | 28 August 2026

Core rule

Group different LLC owners by the same normalized owner tax mailing address — never by the property address.

1. Purpose and intended use

This master specification defines a repeatable process for discovering probable real-estate portfolios held through multiple differently named limited liability companies and producing a controlled direct-mail list. It applies to every metropolitan statistical area (MSA). Each MSA implementation must map its local parcel, owner-mailing, and entity sources to this common model without changing the core grouping or mailing-release rules.

The business question

Which owner tax mailing addresses receive records for two or more distinct LLC owners that collectively hold two or more distinct parcels, and what public entity evidence supports the conclusion that those LLCs may share a common controller or operating portfolio?

Required interpretation

- The result is a probable common-controller or common-portfolio signal, not proof of beneficial ownership.
- A manager, registered agent, organizer, attorney, tax processor, or property manager is not automatically the beneficial owner.
- Every mailed candidate must remain traceable to the public source records and the transformations that created the cluster.

Out of scope

- Filtering to multifamily, 2+ unit, commercial, or any other property class before the LLC-owner test.
- Grouping LLCs by the property or situs address.
- Treating a shared registered-agent address as conclusive proof of shared ownership.
- Naming a person as recipient without credible cross-entity support.

2. Executive requirements

ID	Requirement	Mandatory rule
R-01	Universe	Begin with every parcel in the selected geography.
R-02	LLC test	Keep records whose normalized legal owner name ends with LLC.
R-03	Key	Use normalized owner tax mailing address as the candidate-cluster key.
R-04	Threshold	Require 2+ distinct normalized LLC names and 2+ distinct parcels.
R-05	Evidence	Retain property address as inventory evidence only.
R-06	Review	Flag intermediaries in research; suppress lawyers, agents, CPAs, and service addresses from the automated mailing list.
R-07	Enrichment	Use Secretary of State records after clustering to corroborate entities.
R-08	Outputs	Produce cluster decisions, parcel evidence, a mailing-ready list, and a complete exclusion audit.
R-09	Mail unit	Create exactly one outbound row per normalized owner mailing address.
R-10	Postal gate	Require USPS CASS/DPV and applicable suppression checks before print release.

3. End-to-end workflow

The process separates discovery, qualification, corroboration, and mailing preparation. This prevents entity-enrichment data from biasing the initial portfolio signal.

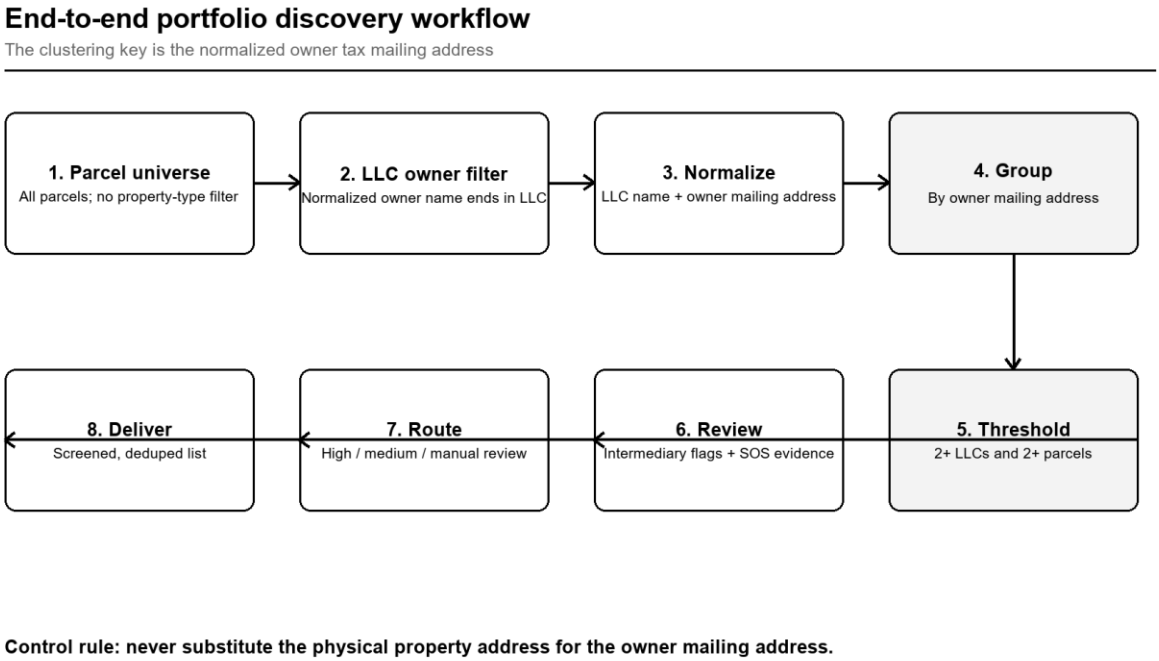


Figure 1. End-to-end portfolio discovery workflow

The owner mailing address is normalized before grouping; Secretary of State evidence enters only after the threshold test.

4. Source data requirements

Minimum parcel-record fields

Field	Meaning	Use	Status
Parcel identifier	Stable parcel/PIN key	Deduplication, evidence, joins	Required
Legal owner name	Tax-roll or assessment owner	LLC suffix filter; entity identity	Required
Owner mailing name	Addressee on tax record	Recipient candidate; QA	Preferred
Owner mailing address	Street, unit, city, state, ZIP	Primary clustering key	Required
Property/situs address	Location of owned parcel	Inventory and evidence only	Required
Source date/version	Assessment roll vintage	Reproducibility	Required

Canonical market-adapter mapping

Canonical field	Local mapping	Processing rule
source_owner_name	Local legal-owner field	Map; normalize; test terminal LLC suffix
source_owner_mailing_name	Local tax-mail addressee field	Recipient evidence and QA
source_owner_mailing_address	Local owner tax-mail address fields	Normalize; primary cluster key
source_property_address	Local situs/property field	Output and evidence only
source_parcel_id	Local PIN/APN/BBL/parcel field	Distinct property count and audit key
source_vintage	Roll year, version, and retrieval date	Reproducibility and change control

5. LLC owner selection

The LLC test applies to the normalized legal owner name. It is intentionally broad and does not depend on property type, unit count, assessed use, or building class.

Required owner-name normalization

- Convert to uppercase and trim leading/trailing whitespace.
- Collapse repeated internal whitespace.
- Standardize terminal variants such as L.L.C., L L C, and punctuation-separated forms to LLC.
- Preserve the original source owner name alongside the normalized value.
- Pass only when the normalized owner name ends with the standalone suffix LLC.

Examples

Source owner name	Normalized owner name	Keep?
Lake View Owner, L.L.C.	LAKE VIEW OWNER LLC	Yes
West Loop Holdings LLC	WEST LOOP HOLDINGS LLC	Yes
LLC Management Services Inc.	LLC MANAGEMENT SERVICES INC	No — does not end in LLC
John and Mary Smith	JOHN AND MARY SMITH	No

6. Correct clustering logic

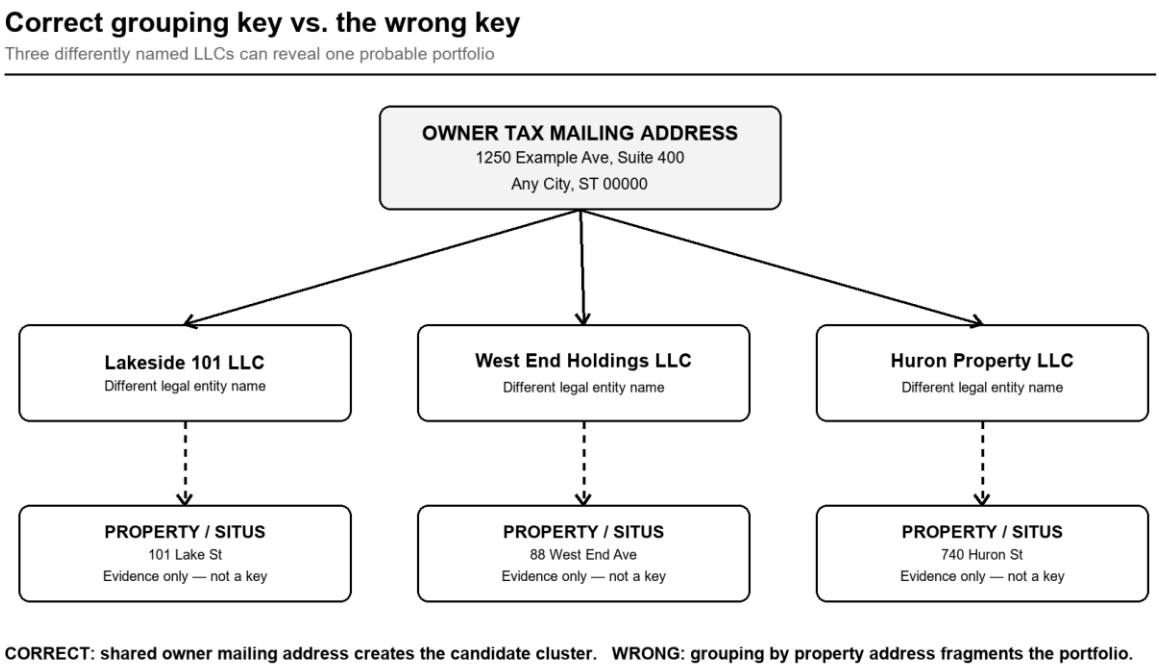


Figure 2. Correct grouping key versus the wrong key

Different property addresses are expected. Their shared owner mailing address is the candidate portfolio signal.

Non-negotiable rule

The physical property address must never be used as the initial grouping key. It identifies the asset, not the place where the owner receives tax mail.

7. Mailing-address normalization

Normalization must improve match consistency without erasing meaningful unit, suite, or postal information. Store both the source value and each normalized component.

Component	Rule
Case and whitespace	Uppercase; trim; collapse repeated spaces
Punctuation	Remove non-semantic punctuation; preserve unit meaning
Street suffixes	Standardize common equivalents, e.g., STREET→ST, AVENUE→AVE
Directionals	Standardize NORTH→N, SOUTH→S, EAST→E, WEST→W when unambiguous
Units	Standardize SUITE/STE and APARTMENT/APT; do not drop the unit
City/state	Normalize spelling and two-letter state abbreviation
ZIP	Use five-digit ZIP as the default; preserve ZIP+4 separately
PO boxes	Standardize PO BOX; retain as a valid mailing address
Missing/invalid	Do not cluster; route to QA/exclusion log

Recommended cluster key

normalized_street_line_1 + normalized_street_line_2 + normalized_city + normalized_state + normalized_zip5

Exact normalized matching is the production baseline. Fuzzy or probabilistic address matching, if later introduced, must be a separate review layer with an explainable score and no silent merges.

8. Data relationship model

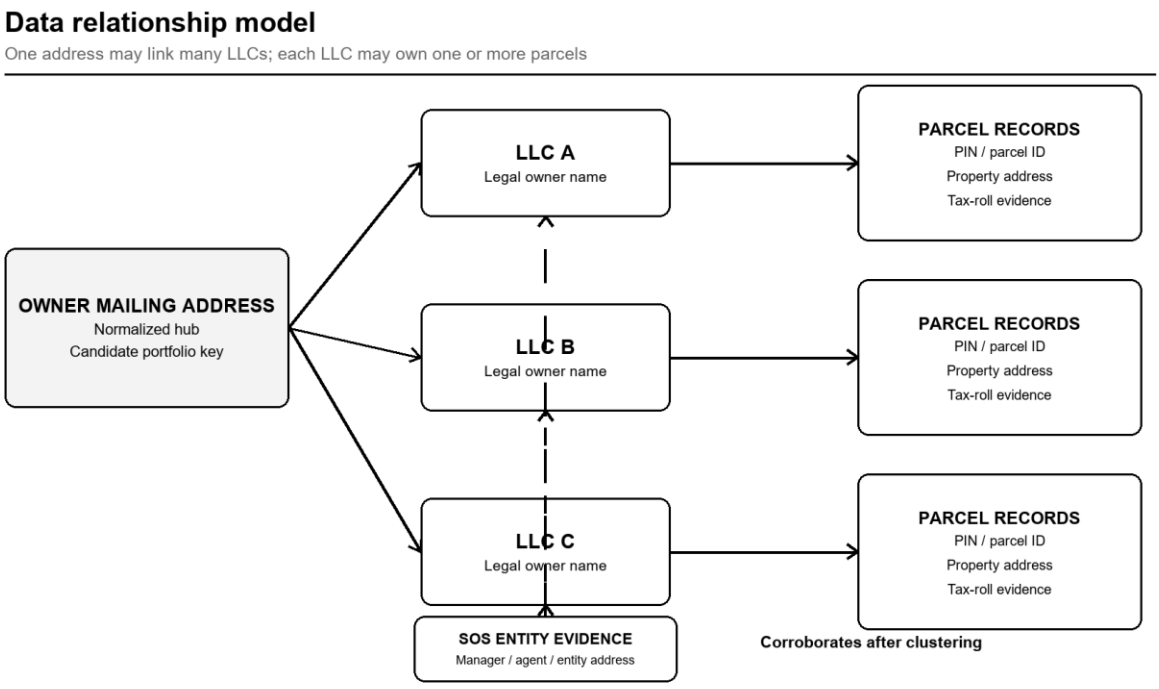


Figure 3. Data relationship model

The address hub creates a candidate cluster; parcel records prove holdings; entity records corroborate after discovery.

9. Cluster construction and qualification

Algorithm

1. Load the complete parcel universe for the target geography and source vintage.
2. Normalize the legal owner name and retain rows where it ends with LLC.
3. Normalize the owner tax mailing address and quarantine missing or invalid values.
4. Group records by the exact normalized owner mailing address.
5. Within each address group, count distinct normalized LLC names and distinct parcel identifiers.
6. Qualify groups with at least two distinct LLC names and at least two distinct parcels.
7. Apply intermediary and mass-service-address flags while retaining the research cluster and its evidence.
8. Enrich each distinct LLC with Secretary of State evidence and assign confidence.
9. Create the cluster summary, property evidence, deduplicated mailing-ready list, and complete exclusion audit.

Formal acceptance expression

normalized_owner_mailing_address → 2+ distinct normalized LLC names → 2+ distinct parcels

Distinctness and deduplication

- A parcel is distinct by stable parcel/PIN identifier, not by formatted property address.
- An LLC is distinct by normalized legal name within the parcel source; entity identifiers may later refine this.
- Duplicate source rows must not inflate parcel or LLC counts.
- One LLC owning several parcels does not satisfy the multi-LLC requirement by itself.
- For outbound mail, all surviving clusters sharing the same normalized mailing address consolidate into one campaign row.

10. Qualification and routing

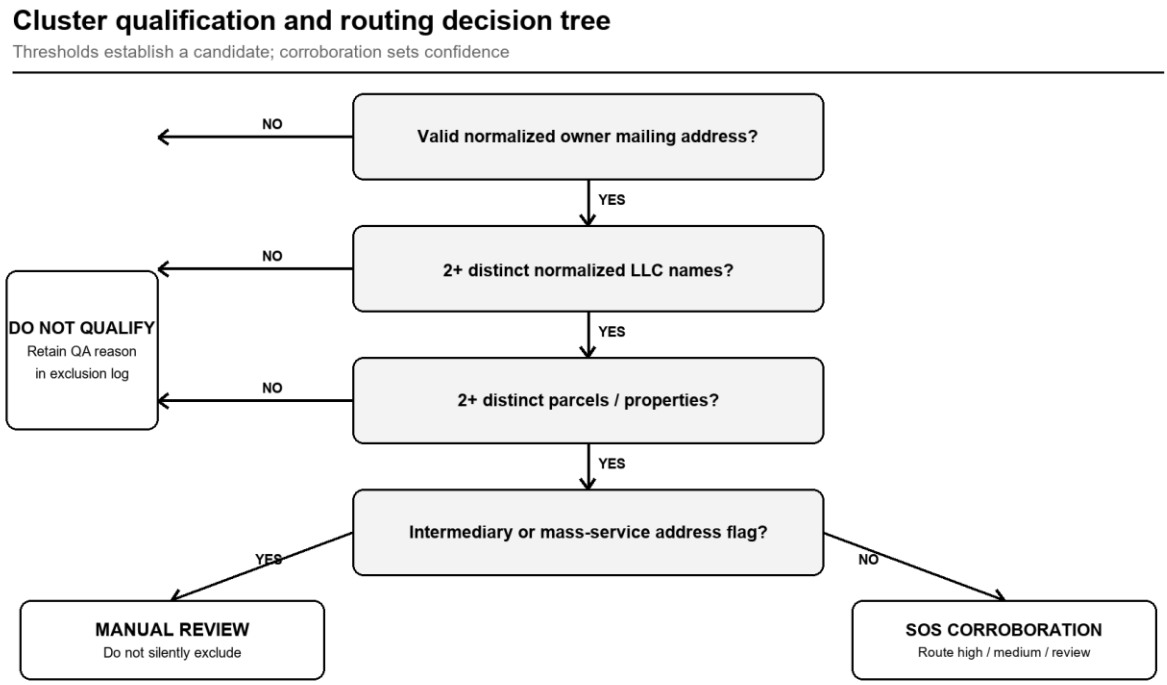


Figure 4. Cluster qualification and routing decision tree

A service-address flag changes the review route; it does not erase the candidate.

11. Intermediary and false-positive controls

Shared addresses may represent a common controller, but they may also belong to professional intermediaries. These cases must be visible, explainable, and reviewable.

Pattern	Observable signal	Required treatment
Registered agent	Address reused across unrelated entities; agent name present	Flag; compare managers/principals and address scale
Law firm	Firm name, attorney suite, many unrelated clients	Flag for manual review
Tax processor/accountant	Tax-service addressee or very high entity volume	Flag; seek corroborating manager evidence
Property manager	Management company receives bills for many owners	Flag; compare owner/entity records
Mass service/virtual office	Address links unusually many LLCs across broad industries	Flag; lower confidence absent corroboration
Property address reused as mail	Mailing address equals one situs address	Flag; do not assume owner residence or control

Scale-based review

The project should calculate the number of distinct LLCs, parcels, owner names, and industry/entity patterns per mailing address. High-volume addresses are not automatically invalid, but require stronger corroboration and often manual review.

12. Secretary of State corroboration

The relevant state entity authority is used after parcel clustering. It should validate entity identity and reveal repeated managers, principals, registered agents, or entity addresses across the LLCs in a candidate cluster. The local market adapter must document the authority, fields, access method, and record vintage.

Recommended entity fields

- Entity file number, exact legal name, status, formation date, jurisdiction, and entity address.
- Registered-agent name and address.
- Manager/member or principal names and addresses where publicly available.
- Annual-report or series-LLC relationships where applicable.

Match hierarchy

1. Exact normalized legal name plus jurisdiction/status.
2. Exact legal name plus corroborating address or file number.
3. Fuzzy name candidates only as manual-review leads; never as silent matches.

Interpretation guardrails

- Manager does not necessarily mean beneficial owner.
- Registered-agent matches alone are weak control evidence.
- Repeated credible manager/principal names or addresses across multiple LLCs materially strengthen the portfolio hypothesis.

13. Confidence framework

Level	Evidence pattern	Routing
High	Shared owner mailing address; repeated credible manager/principal or entity-address support across LLCs; no strong intermediary flag	Eligible for the mailing-ready list after final QA
Medium	Shared owner mailing address; active entity matches; limited direct manager proof; no strong contradiction	Eligible with conservative recipient naming
Low / review	Agent, law firm, CPA/tax processor, property manager, mass-mailing address, situs reuse, ambiguous entity match, or conflicting evidence	Exclude from automated mailing; retain in audit for manual review

Recipient naming

- Use a person's name only when the same credible person is supported across multiple LLCs in the cluster.
- Otherwise use the tax-roll mailing name when suitable.
- If no defensible recipient name exists, use "Property Owner" rather than inventing a person.
- Retain the LLC names and property list in the evidence file even when the outbound card uses a generalized addressee.

14. Required output package

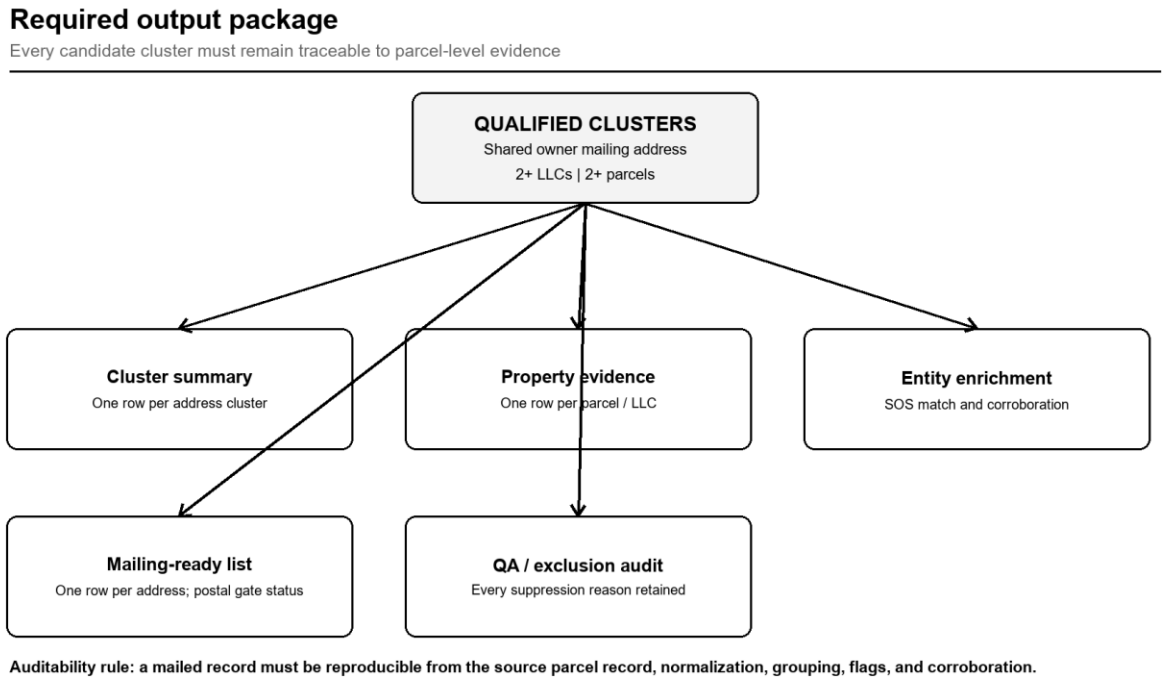


Figure 5. Required output package

The mailing-ready list is a downstream view, not a substitute for cluster and parcel evidence.

15. Output schemas

A. Cluster summary — one row per normalized owner mailing address

Field	Definition
cluster_id	Stable project-generated identifier
normalized_owner_mailing_address	Exact grouping key
source_owner_mailing_addresses	Observed source variants
distinct_llc_count	Count of normalized LLC names
distinct_parcel_count	Count of parcel/PIN values
llc_names	All source and normalized LLC names
property_geography	Cities/neighborhoods/ZIPs represented
intermediary_flags	All triggered review flags
confidence_level	High, medium, or low/review
decision_reason	Human-readable rationale

B. Property evidence — one row per parcel-owner record

Field	Definition
cluster_id	Foreign key to cluster summary
parcel_id / PIN	Stable parcel identifier
source_owner_name	Owner text as published
normalized_llc_name	Normalized distinct-LLC key
property_address	Situs/inventory address; never clustering key
owner_mailing_name	Tax-roll addressee
source_owner_mailing_address	Published mailing address
normalized_owner_mailing_address	Cluster key
source_version	Dataset vintage and retrieval date

C. Entity enrichment and mailing-ready list

Artifact	Minimum fields
Entity enrichment	cluster_id, LLC name, entity/file ID, status, entity address, agent, managers/principals, match method, source date
Mailing-ready list	mailing_id, recipient, full mailing address, parsed postal components, source cluster IDs, distinct LLC/property counts, confidence, exclusion-screen result, postal-validation status
QA/exclusion log	record/cluster ID, target mailing address, rule(s) triggered, evidence detail, source values, reviewer, disposition, timestamp

16. Quality assurance and acceptance tests

Record-level tests

- Every included owner name passes the normalized terminal LLC test.
- Every included parcel has a stable parcel/PIN key.
- Every clustered record has a nonblank normalized owner mailing address.
- Every property address is stored separately from the owner mailing address.

Cluster-level tests

- Every qualified cluster contains 2+ distinct normalized LLC names.
- Every qualified cluster contains 2+ distinct parcel identifiers.
- No duplicate source row inflates either count.
- All intermediary flags are retained with the research cluster.
- Every mailing-ready address appears exactly once.
- No mailing-ready destination is identified as a lawyer, agent, CPA/accountant, filing service, care-of address, or property/situs substitution.
- Confidence and recipient decisions have a written rationale.

Auditability tests

- A reviewer can reproduce the group from the normalized address and property evidence file.
- Original source fields are retained beside normalized fields.
- Dataset vintage, retrieval date, and processing version are recorded.

- Manual overrides identify reviewer, timestamp, prior value, new value, and reason.

Acceptance rule

The process is accepted only when 100% of qualified clusters satisfy the two-LLC/two-parcel threshold, 100% of mailed records link to parcel-level evidence, every outbound normalized mailing address is unique, every screened professional-intermediary record is retained in the exclusion audit, and no property/situs address is used as a clustering key.

17. Market adaptation checklist

1. Define the target geography and obtain the complete parcel universe.
2. Confirm which source fields contain legal owner name, owner mailing name/address, property address, and parcel ID.
3. Document access method, terms, refresh cadence, and dataset vintage.
4. Map local LLC suffix variants and address conventions into the normalization rules.
5. Locate the relevant Secretary of State entity and manager/agent records.
6. Profile high-volume mailing addresses and build local intermediary indicators.
7. Run threshold and QA tests before entity enrichment or mailing selection.
8. Perform a documented sample review across high-, medium-, and review-routed clusters.
9. Freeze the output schema, processing version, and source snapshot for reproducibility.

Market configuration record

Configuration item	Required entry
Market / MSA	Name and geographic boundaries
Parcel authority	Agency, dataset, access method, refresh date
Entity authority	Secretary of State source and refresh date
Field mapping	Local field → canonical field
Normalization version	Named ruleset/version
Thresholds	Default: 2+ LLCs and 2+ parcels
Known intermediaries	Local firms, agents, processors, virtual offices
QA owner	Reviewer and approval date

18. Mailing preparation and operating safeguards

Mailing preparation begins only after cluster qualification, entity review, and recipient routing. The research dataset and the outbound mailing list should remain separate artifacts so later suppression or compliance decisions do not alter the underlying evidence.

Mandatory mailing-list suppression and deduplication

Control	Mandatory treatment
Duplicate address	Normalize the complete owner mailing address, preserving suite/unit; consolidate to one outbound row and one card per resulting key.
Lawyer / law firm	Exclude when the target destination or supported addressee is legal counsel, an attorney, law office, or law firm.
CPA / accountant	Exclude when the target destination or supported addressee is a CPA, accountant, bookkeeping firm, or tax-preparation service.
Registered/process agent	Exclude when the target destination matches a registered-agent address, process-agent address, filing service, corporation service, or known mass-agent location.
Care-of destination	Exclude C/O and care-of targets from the automated campaign; retain for manual review.
Property-address substitution	Exclude when the proposed owner mailing address is actually a property/situs address marker.
Low/manual-review confidence	Exclude from automated release and retain the reasons and evidence in the audit log.

- Validate address deliverability and standardization through an authorized postal process before printing.
- Maintain suppression lists for returned mail, opt-outs, do-not-contact decisions, and duplicate households/addresses.
- Do not include sensitive personal information on the card or in exposed tracking parameters.
- Use public-record information only for the documented business purpose and review applicable federal, state, and local solicitation rules before launch.
- Retain the research basis, mailing version, and campaign date so each card can be traced to an approved mailing-ready record.

Pre-mail release gate

Gate	Required approval
Data	Cluster and parcel evidence passed all acceptance tests
Identity	Recipient name and confidence treatment follow this specification
Deduplication	Exactly one campaign row exists for each normalized owner mailing address
Professional suppression	Lawyers, agents, CPAs/accountants, filing services, care-of addresses, and flagged situs substitutions are absent from the outbound list and present in the exclusion audit
Postal	USPS CASS/DPV completed; NCOA and campaign suppressions applied when appropriate
Content	Card copy approved; no unsupported ownership claims
Compliance	Campaign reviewed for applicable solicitation/privacy requirements
Archive	Source snapshot, mailing version, exclusion audit, and approval log retained

19. New-MSA implementation baseline

Every MSA is implemented as a market adapter under this master specification. Local source names and availability may differ, but the canonical fields, clustering rule, qualification thresholds, mailing suppressions, deduplication rule, evidence package, and release gates remain common.

Implementation sequence

1. Create the MSA folder below this master specification.
2. Define the geographic boundary and acquire a versioned parcel/assessment universe.
3. Map local legal-owner, owner-mailing, situs, parcel-ID, and source-vintage fields to the canonical model.
4. Apply the terminal LLC test to all properties without a property-class or unit-count prefilter.
5. Normalize the owner mailing address, group by that key, and apply the two-LLC/two-parcel threshold.
6. Profile and flag professional intermediaries, service addresses, care-of destinations, and property-address substitutions.
7. Match qualifying LLCs to the relevant state entity authority after clustering.
8. Publish cluster evidence, property evidence, a deduplicated and screened mailing file, and the exclusion audit.
9. Complete postal validation, compliance review, and print authorization.

MSA activation requirement

An MSA is ready to trigger only when its source snapshot, canonical field mapping, normalization version, intermediary-screening method, entity-source access, output locations, QA reviewer, and postal-validation handoff are documented and testable against this master specification.

20. Definitions and version control

Term	Definition
Owner mailing address	Address where the parcel authority sends owner/tax correspondence; the clustering key
Property/situs address	Physical location of the parcel; inventory evidence only
Candidate cluster	Records sharing one normalized owner mailing address before final confidence review

Term	Definition
Qualified cluster	Candidate meeting 2+ LLC and 2+ parcel thresholds
Intermediary flag	Signal that the address may belong to a professional service provider
Corroboration	Independent entity evidence supporting a common-control hypothesis
Probable portfolio	Qualified and reviewed cluster; not a legal conclusion of beneficial ownership

Document control

Version	Date	Status	Change
2.0	28 Aug 2026	Master	Generic all-MSA specification; one-piece-per-address deduplication; lawyer, agent, CPA/accounting, filing-service, care-of, situs, and confidence suppressions; postal release gate

Future changes should preserve the core grouping rule unless a formally approved version explicitly replaces it: group different LLC owners by normalized owner tax mailing address, not by property address.